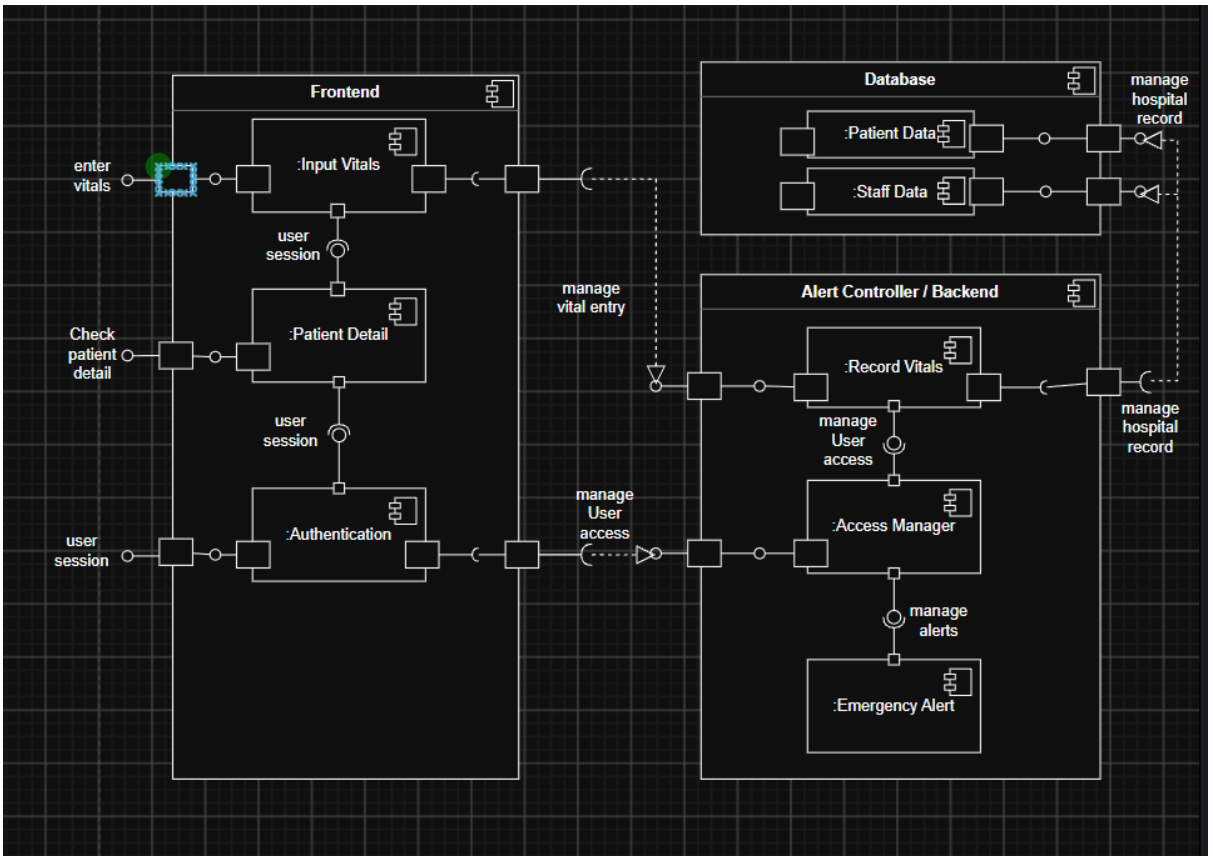
PS TITLE -> Remote Patient Vitals Alert & Monitoring App

UML COMPONENT Diagram ->



The UML Component Diagram for the Remote Patient Vitals Alert & Monitoring App represents the major software components of the system and the interactions between them. The system is designed to allow patient vital information to be entered and monitored remotely, while providing secure access to patient records and generating alerts when necessary.

The architecture is divided into three major sections: Frontend, Alert Controller/Backend, and Database. The Frontend provides the user-facing interface through which users interact with the application. It consists of three main components: Input Vitals, Patient Detail, and Authentication. The Input Vitals component allows authorized users to enter patient vital information. The Patient Detail component is responsible for viewing and checking patient information, while the Authentication component manages user sessions and ensures that only authenticated users can access protected functionality.

The Alert Controller/Backend acts as the central processing layer of the application. It contains the Record Vitals, Access Manager, and Emergency Alert components. The Record Vitals component receives and manages vital entries submitted through the frontend and communicates with the database to store or retrieve relevant information. The Access Manager controls user permissions and determines whether a user is authorized to access particular patient or hospital records. The Emergency Alert component is responsible for

managing alerts when monitored vital information requires attention. This component can notify the appropriate users or staff when an abnormal or critical condition is detected.

The Database layer provides persistent storage for the application's information. It contains separate components for Patient Data and Staff Data. Patient Data stores patient-related information and vital records, while Staff Data maintains information about authorized healthcare personnel. The backend communicates with these database components to retrieve and update records.

The diagram also shows the flow of important interactions between the components. Users provide their vital entries through the frontend, which are passed to the backend for processing and storage. Authentication establishes the user session, while the Access Manager handles user access and permissions. The backend can manage hospital records and retrieve patient information from the database. When required, the processed vital information is passed to the Emergency Alert component, which manages alerts for potentially critical situations.

Overall, the component diagram demonstrates a layered architecture in which the frontend handles user interaction, the backend performs processing, access control and alert management, and the database provides persistent storage. This separation makes the application easier to maintain, secure, and extend while supporting the core requirement of **r**emote patient vital monitoring and timely alert generation.